GHG Statement Report

Version 1.6

Information requirements for verification are as follows:

- GHG Statement Report (this document) containing information relating to the GHG Statement
- GHG Statement calculations (provided separately)
- Supporting information including copies of raw data used

An appropriate level of information should be provided with a clearly referenced and transparent audit trail of decision making, assumptions, explanations, such that a verifier can trace all inputs, outputs and decision making. All data sources and assumptions must be clearly referenced, transparent and traceable.

After this document has been completed for the first time, it may be acceptable to refer back to relevant sections for subsequent Reporting Periods, for example if there have been no changes to the GHG Assessment scope, data collection methodology, calculation methodology or assumptions.

There is no word limit for answers in sections below.

Section A – General Project Information

General Information	
Name of practitioner who prepared the GHG Statement, and relevant competencies. This should include an overview of relevant qualifications and experience undertaking GHG assessments.	Max Lavine Charm Industrial Protocol and Verification Lead
Name of Project / link to Project page on Isometric Registry	
Date of report	5/13/25
Please specify the Reporting Period this GHG Statement has been prepared for (the Reporting Period describes the time period over which the carbon removal activity assessed in the GHG Statement occurred)	2/1/25-3/31/25
Relevant project information if not included in PDD	

Details and links to any supporting documentation	All raw data and supporting documentation has been submitted either 1) directly to Isometric via API, or 2) directly to the VVB
Please confirm that the GHG Statement has been undertaken in accordance with the information provided in the project boundary section of the PDD. Provide any additional information relating to the GHG Statement project boundary here.	Confirmed

Section B – Project Details

B. GHG Statement - Project Details
Provide information on the Project design and implementation concerning the reporting period. *This should include the following information as a minimum:* ● *Key features of the Project including a technical description* ● *information on the implementation status of the project, For the description of the installed technologies, technical processes and equipment, include diagrams, where appropriate*
Updated Embodied Emissions Inventory The embodied emissions inventory for the project has been updated to reflect the installation of new equipment and the completion of capital projects that were in progress during previous reporting periods. This includes the completion of a detention pond infrastructure project in Colorado and site electricity upgrades in Kansas. ### Updated Emission Factors March of 2025 marks a year since the initiation of the project. As part of that milestone, we have updated our project emission factors to the most current available version from all relevant databases, sources, etc. While this reporting period also includes February injections, all non-production activities' emissions are estimated using the updated factors so all removals could be reported in a single batch. Because production activities' emissions are separately calculated using spreadsheets, only March production is estimated using updated factors. Production during December, January, and February are estimated using the same emission factors as used during previous periods. Emission factors are not updated for estimates relating to emissive activities that occurred in the past such as embodied emissions or the emissions created from transporting biomass that has been stockpiled and used progressively over time.

Section C – Project Changes

C.1 Summary of changes

Please provide details of any changes or deviations from the planned implementation, operation, or monitoring of the project activity that occurred during this reporting period. Project Proponents must determine whether the changes are temporary deviations or permanent changes.
Temporary deviations could include deviations from:

a) The monitoring plan mentioned in the approved PDD.
b) The applied Protocol.
c) The applied modules.
d) Data collection procedure.

If there were any temporary changes or deviations corresponding to this reporting period, please include the details of those changes in this section. If the changes do not fall into the above categories, please note them in Section F of the corresponding PDD instead.

(If there were no deviations during the reporting period, this section is not applicable. In that case, kindly leave this section blank and mention "Section is intentionally left blank.")

Charm-Produced Bio-Oil Injections – Production Period MRV

Charm's tech team is in the process of building out Ledger functionality that will allow for the bulk transmission of datapoints associated with bio-oil production in CO to Isometric's DMRV. In the interest of expediency, we are verifying the removals included in this GHG Statement before that work is finished.

In order to fully account for bio-oil production emissions, Charm's MRV team has collected and calculated the datapoints using spreadsheets. Individual calculation sheets, provided separately to the VVB as part of Charm's evidence package, are rolled up into a master sheet which is provided as evidence for Charm CO Production Emissions in Isometric's DMRV. This sheet includes emissions from:

1. Biomass transport
2. Propane combustion
3. Diesel combustion
4. Gasoline combustion
5. Electricity usage
6. Tailgas
7. Water usage
8. Consumables delivery
9. Waste disposal
10. Amortized embodied emissions from equipment and capital goods
11. Embodied emissions from spare parts and consumables

These emissions are summed and allocated to pyrolysis products by proportion of carbon mass. Prior to allocation:

1. Measurement uncertainty is manually added to all relevant parameters
    a. When fully integrated, the system provides uncertainty data as a standard deviation alongside relevant datapoints. The standard deviation is used as an input for variance propagation in Isometric's DMRV which computes an uncertainty discount
        i. This is the method for uncertainty discounting used for all other Charm removals and processes
    b. Because that is not yet feasible for Charm's bio-oil production, uncertainty for those datapoints is addressed using Option A (Conservative estimate of input parameters) rather than Option B (variance propagation). That is achieved by adjusting the relevant input parameter to reflect the maximum possible level of measurement uncertainty
        i. Ex. 1: The maximum allowable error for a legal-for-trade truck scale in the US is +/- 160 lbs. Charm receives a BOL for biomass reflecting a 10,000 lb load. A total mass of 10,160 is used as an input to scale the associated biomass transport emissions.
        ii. Ex. 2: The maximum allowable error for an electricity meter provided a utility in CO is +/- 2%. Charm notes that 64,000 kWh have been used in a production period. The associated electricity generation emissions are scaled by 64,000*(1+.02) or 65,280 kWh
2. Procured renewables are applied to the parameter where applicable
    a. Charm's inventory of both BCUs and RECs is tracked in inventory management spreadsheets shared with Isometric. The use of these procured renewables is documented in the production period summary spreadsheet. The inventory tracking spreadsheet is adjusted accordingly to ensure that procured renewables are not double-counted

Charm-Produced Bio-Oil Injections – C Content Sampling

1. Charm bio-oil is injected as a blend with AECN bio-oil. For this reporting period, C content of the blended batches is calculated using Method A (measure every batch) option 2: Sampling the constituent production batches to calculate a linear weighted average C content for the injection batch
    a. The above applies to injection batches 3-07 through 3-11. For batches 3-12 on, the sample analyzed for C content was sampled directly from the batch after AECN and Charm oil had been fully homogenized. So, that sample is representative of the full batch and no weighted average is necessary to compute the C content

RECs and BCUs Accounting

1. Charm's tech team is also in the process of building support for the use of RECs as specific components in DMRV. While that is in progress, grid electricity emissions are shown as electricity emissions with an emission

factor representative of the emission factor associated with the generation method represented by the RECs

   a. Because 100% of energy consumption is volumetrically matched by RECs, there are not electricity emissions that are scaled by the emission factor for overall generation in the grid region

2. Because Charm bio-oil production emissions are calculated outside of Ledger, the BCUs and RECs used in the Charm bio-oil emissions accounting also do not show up as separate components in Isometric DMRV. Rather their impact is reflected in the total carbon intensity of Charm oil used in removals
3. For full clarity and transparency, a total of 288.42 MT CO2e in emissions reductions from BCUs and a total of 259.38 MWH of RECs were used during this reporting period

C.2 Impact of changes

For each deviation, provide a description of the nature, extent and duration of the non-conforming reporting period, and:

a) Propose alternative monitoring arrangements for the non-conforming reporting period; or

b) Apply the most conservative values approach

(If there were no deviations during the reporting period, this section is not applicable. In that case, kindly leave this section blank and mention "Section is intentionally left blank.")

Table: 1 Temporary changes to the project

Period of change	*Nature of change*	*Impact of change*
Current Reporting Period	Use of spreadsheets, rather than Ledger API, to compute carbon intensity of Charm-produced bio-oil	De Minimis
Current Reporting Period	Use of weighted average C content of batch constituents to compute C content for Charm-blended batches	De Minimis
Current Reporting Period	Use of renewable generation CI for electricity consumption matched by RECs	De Minimis

Section D – GHG Statement Methodology

D.1 GHG Statement Methodology - Data
Provide information on the GHG Statement approach and methodology in relation to data collection. *This should include the following information as a minimum:* ● *Data collection procedures - how data was requested and gathered, who supplied each data point, the format data was received in, and the date data points were received* ● *How data quality was assessed and how the data hierarchy was followed in terms of using measured, calculated and estimated data. Evidence and justification should be provided in any cases where measured data was not used* ● *Details of data validation processes followed, including treatment of missing data* *Evidence of all raw data that informed the assessment (including data that informed any assumptions) must be appended to the PDD, or linked within the GHG Statement.*
See corresponding section of PDD

D.2 GHG Statement Methodology - Calculations
Provide information on the GHG Statement approach and methodology in relation to calculations. *This should include the following information as a minimum:* ● *General description of the methodology, criteria and procedures used as a basis for the assessment, including reference to any documentation (including Isometric protocols and modules), guidance, industry standards and best practice that were followed* ● *Information on calculation procedures followed* ● *Information on any tools used as part of the assessment*

● *Procedure for selecting emission factors including how age (age of data, and the period over which they have been collected), geography (the region or country from where the data have originated), technology (whether the data are specific to a particular technology or mix of many), methodology (the approach applied to gather or calculate the data) and competency (proficiency of entity that developed the data) were considered.* ● *Assumptions and limitations - provide full transparency in terms of value-choices, rationales and expert judgements* ● *Details of a sensitivity analysis, how parameters were assessed and evidence behind choices* ● *Details of uncertainty analysis* ● *Details of any uncertainty adjustments (e.g. %) applied in instances of high uncertainty*
See corresponding section of PDD

Section E – GHG Statement Results

GHG Statement Results	
The following information should be provided in this section: ● *Baseline results reported in t CO_2e for the Reporting Period* ● *Net CO_2e removals results reported in t CO_2e for the Reporting Period. These must be aggregated and also broken down into GHG SSRs* ● *If it has not already been covered in the* 'Uncertainty	

assessment' section of the PDD, provide *outcomes of sensitivity and uncertainty analyses including a statement of how uncertainty affects the results and how it has been addressed to minimize misrepresentation* ● *A statement of how the assumptions and choices made in the assessment are conservative* *The GHG Statement calculations must be provided separately, including all raw data and evidence. The GHG Statement calculations must be clearly referenced with a transparent audit trail of evidence, decision making, assumptions, explanations.*		
	Kg	**MT**
Gross Bio Oil Sequestered	764318	764.318
	Kg CO2e	**MT CO2e**
Gross CO2e Sequestered	1224121	1224.121
Project Emissions	**Kg CO2e**	**MT CO2e**
Process Emissions	390136.3677	390.136
Site+Amortized Embodied Emissions	52223.6323	52.224
Uncertainty Discounts	3303	3.303
Emissions Reductions	224492	224.492
Total Emissions for Reporting Period (kg)	445663	
Total Emissions for Reporting Period (MT)	445.663	
Net Removals for Reporting Period (kg)	1002950	
Net Removals for Reporting Period (MT)	1002.95	

Section F – GHG Statement Version Changes

GHG Statement Version Changes
For each statement resubmission, please specify the new version date (which is the same as the resubmission date) and the updated total of CO_2e tonnes removed.

GHG Statement Versions

Version date	*Total CO_2e tonnes removed*
5/13/2025	1002.95

Document History

Version	**Date**	**Description**
v1.6	June 2024	Revision to include Section F - GHG Statement Version changes, in line with the versioning product experience
v1.5	May 2024	Revision to include section on project changes, and other editorial changes
v1.4	November 2023	Public version